

Qatar-2b

R-1737

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_220306 · created 2026-08-02T18:13:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459644.9787 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.056
Transit depth (Rp/Rs)^2	5.2 +/- 2.55 [%]
Orbital Inclination (inc)	89.64 +/- 1.51
Airmass coefficient 1 (a1)	0.64 +/- 0.22
Airmass coefficient 2 (a2)	0.33 +/- 0.14
Scatter in the residuals of the lightcurve fit is	32.78 %
Best Comparison Star	#1 - [291, 429]
Optimal Aperture	3.3
Optimal Annulus	8.76
Transit Duration (day)	0.0808 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.228 $\pm$ 0.056 vs 0.16327 (deviation 0.39 σ)
Duration fit vs published	0.0808 d vs 0.0767 d (deviation 0.35 σ)
Mid-time O-C	3.1 min
Depth SNR	1.4
Residual scatter	32.78 %

Light curve

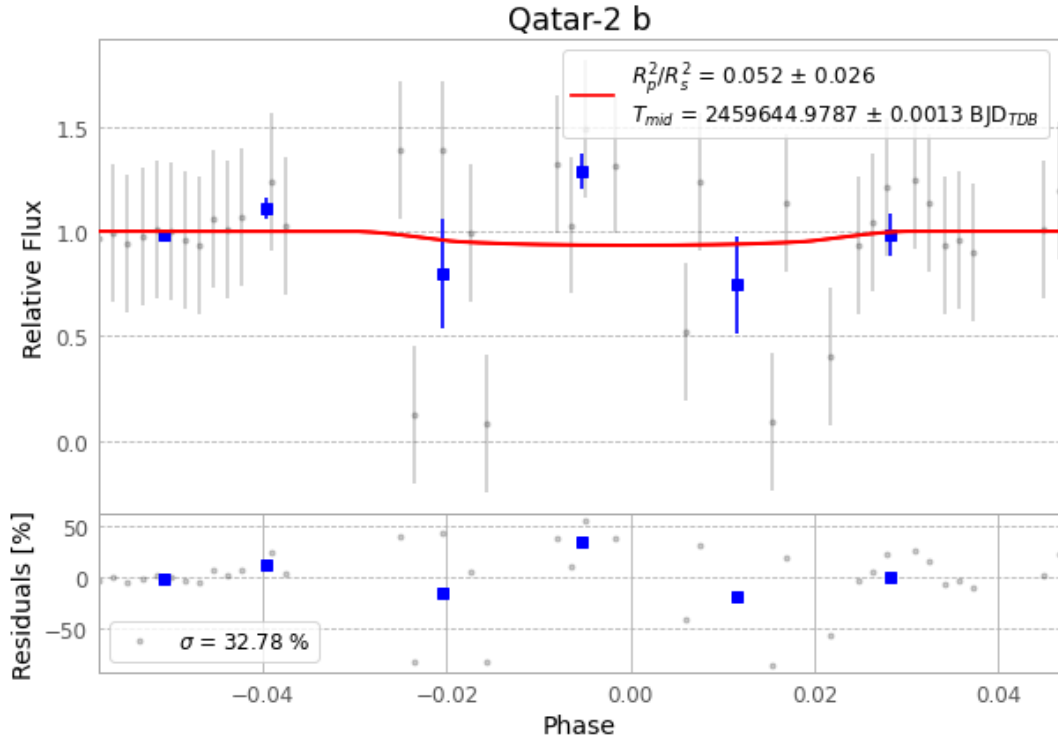


Figure 1 — FinalLightCurve_Qatar-2 b_2022-03-06.png (SHA-256 a19ad72fb8f575f3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 461], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5693db3c260d0507...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), Qatar-2220306093012.FITS → Qatar-2220306125112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-220306.log (1051745e427e...), FinalLightCurve_Qatar-2 b_2022-03-06.png (a19ad72fb8f5...), FinalLightCurve_Qatar-2 b_2022-03-06.csv (105649eff8e7...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:13Z.